

Synthesize the literature needed to introduce artificial intelligence to non-specialist readers in an industrial design dissertation focused on parametric customization and assistive product development. Explain clearly what artificial intelligence is, how it works in basic terms, and how it differs from machine learning, deep learning, and generative AI. Identify the forms of AI most relevant to design, especially decision support, optimization, computer vision, natural language processing, and generative systems. Analyze how AI is used across the design process, including concept development, exploration of alternatives, personalization, evaluation, and support for project decisions. Discuss the role of the designer in relation to AI, emphasizing that AI functions as an assistive tool rather than an autonomous substitute. Also address limitations and risks such as bias, opacity, error, overreliance, and the need for human oversight. Prioritize literature suitable for writing an accessible but academically rigorous introductory section for readers with no technical background

The literature consistently establishes that AI—spanning decision support, optimization, computer vision, natural language processing, and generative systems—augments rather than replaces the industrial designer across all process stages, serving as a powerful assistive tool for parametric customization and assistive product development while requiring sustained human oversight to manage inherent risks of bias, opacity, error, and creative homogenization.

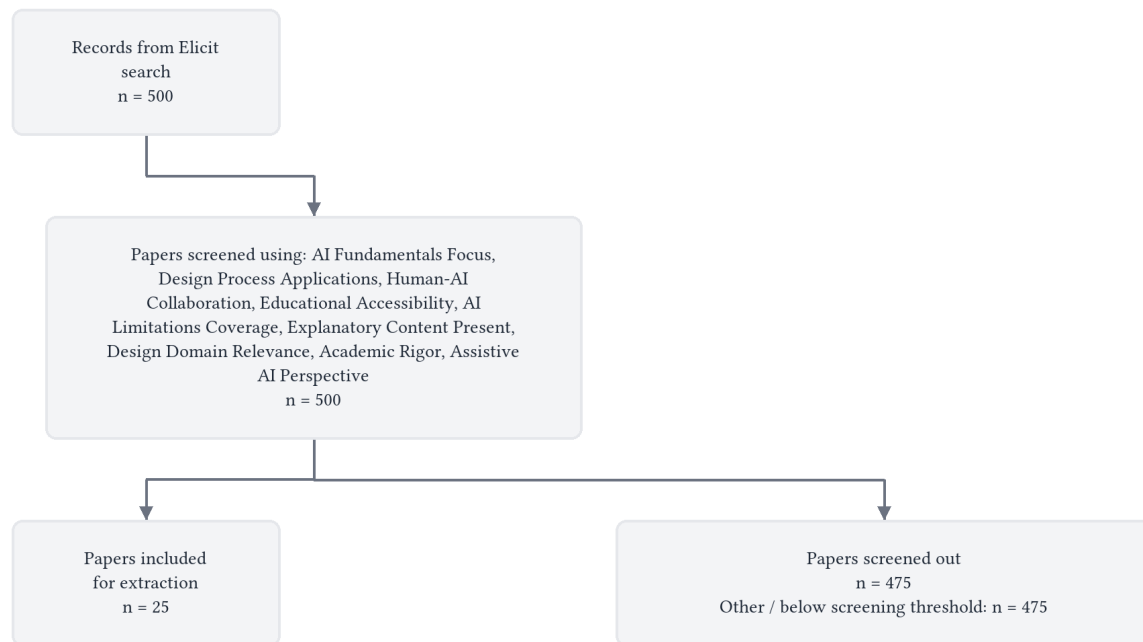
Abstract

Artificial intelligence is best understood as a nested hierarchy: AI is the umbrella term for computational techniques that approximate cognitive functions such as learning and pattern recognition; machine learning is the sub-discipline that trains algorithms from examples; deep learning uses artificial neural networks to mimic brain-like learning; and generative AI applies deep learning to produce novel creative outputs from user prompts [1, 2]. Five forms of AI are most relevant to design practice—decision support, optimization, computer vision, natural language processing, and generative systems—with generative tools (e.g., DALL-E, Midjourney, Stable Diffusion) dominating recent attention [2, 3], while optimization and parametric systems (e.g., Finch 3D, Autodesk Generative Design) and NLP-based interfaces are particularly salient for parametric customization and assistive product development [4–6]. AI contributions map onto recognized design-process stages: broadening ideation through rapid variant generation [7, 8], supporting problem framing through data analysis [9, 10], accelerating prototyping through simulation and sketch-to-mockup translation [4, 11], and enhancing evaluation through performance prediction and user testing [3, 12]. AI is consistently strongest in divergent phases, while convergent selection and contextual interpretation require human judgment [1, 8].

The literature is unanimous that AI functions as an assistive tool rather than an autonomous substitute for the designer, whose irreducible contributions include problem definition, cultural sensitivity, ethical reasoning, and the integration of tacit knowledge that AI cannot access [3, 7, 10, 13]. Documented risks include bias from skewed

training data [2, 8], opacity of neural-network decision-making [1, 5], hallucination and unreliable outputs [1, 11], creative atrophy and stylistic homogenization from overreliance [8, 14], and unresolved questions of authorship and originality [2, 7]. Mitigation strategies include deploying AI in divergent exploration while retaining human veto in convergent selection [8], structured role-reflection protocols that reward deliberate non-use [11], and cultivating AI literacy that encompasses technical competence, ethical reasoning, and critical evaluation [15]. For parametric customization and assistive product development, the evidence supports a layered integration in which optimization systems handle constraint-based exploration, NLP interfaces lower barriers for non-specialist users, generative tools accelerate visual communication, and the designer retains strategic and ethical oversight throughout [3, 4, 6, 16].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Synthesize the literature needed to introduce artificial intelligence to non-specialist readers in an industrial design dissertation focused on parametric customization and assistive product development. Explain clearly what artificial intelligence is, how it works in basic terms, and how it differs from machine learning, deep learning, and generative AI. Identify the forms of AI most relevant to design, especially decision support, optimization, computer vision, natural language processing, and generative systems. Analyze how AI is used across the design process, including concept development, exploration of alternatives, personalization, evaluation, and support

for project decisions. Discuss the role of the designer in relation to AI, emphasizing that AI functions as an assistive tool rather than an autonomous substitute. Also address limitations and risks such as bias, opacity, error, overreliance, and the need for human oversight. Prioritize literature suitable for writing an accessible but academically rigorous introductory section for readers with no technical background”

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **AI Fundamentals Focus:** Does the study focus on AI fundamentals and definitions, explaining what artificial intelligence is, its basic mechanisms, or distinguishing between AI, machine learning, deep learning, and generative AI in accessible terms?
- **Design Process Applications:** Does the study examine AI applications in design processes such as concept development, alternative exploration, personalization, evaluation, or decision support within design contexts?
- **Human-AI Collaboration:** Does the study examine human-AI collaboration in design, exploring the designer's role when working with AI tools and emphasizing AI as assistive rather than autonomous?
- **Educational Accessibility:** Is the publication written for non-technical audiences or does it contain clear explanatory components suitable for teaching AI concepts to design practitioners?
- **AI Limitations Coverage:** Does the study address AI limitations and risks such as bias, opacity, errors, overreliance, or the need for human oversight in AI systems, particularly in design contexts?
- **Explanatory Content Present:** Does the study include accessible explanations rather than focusing solely on complex algorithms, mathematical proofs, or technical implementation details without explanatory content?
- **Design Domain Relevance:** Does the study focus on AI applications within design domains, or if examining other fields (medicine, finance, autonomous vehicles), are the findings directly applicable to design principles?
- **Academic Rigor:** Is this an academic publication with peer review rather than commercial materials, industry white papers, marketing materials, or non-peer-reviewed content?
- **Assistive AI Perspective:** Does the study promote or examine AI as an assistive tool for designers rather than promoting fully autonomous design systems that replace human designers?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 475

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **AI Definitions:**
Extract all definitions, explanations, and conceptual frameworks for artificial intelligence, machine learning, deep learning, and generative AI that are suitable for non-specialist readers. Include:
 - Basic explanations of what AI is and how it works
 - Distinctions between AI, machine learning, deep learning, and generative AI

- Conceptual frameworks or models used to explain AI concepts
- Any analogies, metaphors, or simplified explanations
- Technical concepts explained in accessible language

- **Design-Relevant AI Types:**

Extract information about specific types of AI applications most relevant to design practice, including:

- Decision support systems and their design applications
- Optimization algorithms in design contexts
- Computer vision applications for design
- Natural language processing in design workflows
- Generative systems (text, image, 3D, etc.)
- Any other AI technologies specifically applied to design
- Capabilities and limitations of each type in design contexts

- **AI in Design Process:**

Extract detailed information about how AI is integrated across different stages of the design process, including:

- Concept development and ideation support
- Exploration of design alternatives and variations
- Personalization and customization capabilities
- Design evaluation and testing assistance
- Decision support for design choices
- Specific examples of AI tools or methods at each stage
- Evidence of effectiveness or impact on design outcomes

- **Designer-AI Relationship:**

Extract evidence and discussion about the relationship between designers and AI systems, specifically focusing on:

- AI as assistive tool vs. autonomous replacement
- Human-AI collaboration models and workflows
- Designer agency and control in AI-assisted processes
- Skills or competencies needed for designers to work with AI
- How AI augments rather than replaces human creativity and judgment
- Examples of successful human-AI collaboration in design

- **AI Limitations & Risks:**

Extract comprehensive information about limitations, challenges, and risks of AI in design, including:

- Bias issues and their manifestations in design AI
- Opacity and explainability challenges
- Error rates, failure modes, or unreliable outputs
- Risks of overreliance or inappropriate dependence on AI
- Need for human oversight and validation
- Ethical concerns specific to design applications
- Mitigation strategies or best practices for managing risks

- **Parametric & Assistive Relevance:**

Extract any content specifically relevant to parametric customization and assistive product development, including:

- AI applications in parametric design or mass customization
- Use of AI in assistive technology development
- Personalization algorithms for individual user needs
- AI-driven adaptation or optimization for accessibility
- Examples from assistive product design or parametric systems
- User-centered AI applications in these domains

- **Educational Content:**

Extract content specifically designed for educational purposes or non-specialist audiences, including:

- Pedagogical approaches to teaching AI concepts
- Learning frameworks or curricula for AI literacy
- Recommendations for introducing AI to design students or practitioners
- Educational examples or case studies
- Guidance for educators on AI integration
- Content explicitly designed for accessibility to non-technical readers

Results

Characteristics of Included Studies

The corpus comprises 25 sources spanning 2018–2026, drawn from architecture, graphic design, product and industrial design, textile design, interior design, engineering education, and human-computer interaction. The majority are qualitative reviews or conceptual papers; several include empirical components such as educational experiments, student reflections, or case studies. The table below summarizes each source.

Study	Full text retrieved?	Study type	Design domain	Primary focus
Amin Mirzaei et al., 2026	Yes	Systematic literature review (205 studies) [8]	Design thinking (cross-domain) [8]	AI across design thinking stages [8]
Salvatore Di Dio et al., 2024	Yes	Narrative review [17]	Product, graphic, service, speculative design [17]	AI applications, ethics, and designer roles [17]
Saja A. Hafez, 2025	Yes	Systematic review (2019–2024) [18]	Graphic design [18]	Hybrid intelligence framework for graphic design [18]
Kristin A. Bartlett & J. Camba, 2024	Yes	Discussion / position paper [2]	Product design education [2]	Ethical and pedagogical implications of image-generative AI [2]

Study	Full text retrieved?	Study type	Design domain	Primary focus
T. H. Teo & Dexter X. H. Chew, 2026	Yes	Literature review [15]	Design intelligence (cross-domain) [15]	AI literacy framework for design [15]
K. Thoring et al., 2023	Yes	Research agenda / conceptual paper [1]	Design (general) [1]	Research agenda for generative AI-enabled design [1]
Y. Lai et al., 2023	Abstract only	Qualitative study (interviews, focus groups) [9]	Design process (general) [9]	GAI impact on double-diamond design process [9]
Benjamin D. B. Klein et al., 2026	Abstract only	Survey / review [19]	3D CAD / engineering design [19]	Generative AI for CAD and designer intelligence [19]
Xiang Liu, 2026	Yes	Systematic review (PRISMA, 59 articles) [3]	Architecture [3]	Architect role transformation under AI [3]
J. S. Wong, 2018	Abstract only	Essay / commentary [20]	Interaction design [20]	Design fiction and civic AI [20]
Züleyha Değirmenci, 2024	Yes	Mini review [21]	Textile design [21]	AI's impact on creativity in textiles [21]
Samuelle Bourgault & J. E., 2023	Yes	Editorial with case studies [6]	Design, craft, HCI [6]	Computation for creativity, CADgpt tool [6]
Olena Shabanova, 2025	Abstract only	Review / analysis [14]	Interior design [14]	AI risks, opportunities, and authorial style [14]
Chang Xiong et al., 2026	Yes	Empirical case study (observation, interviews) [16]	Interface / graphic design [16]	Human-AI collaboration in GenAI workflows [16]
R. H. Wardi et al., 2025	Yes	Concept paper [13]	Crowdsourced design [13]	Reciprocal human-AI framework [13]
A. Troshkin & D. Makukhin, 2025	Yes	Educational experiment + comparative analysis [4]	Architecture and design [4]	AI tools, vibe-coding, and educational transformation [4]
Mirko Stanimirović et al., 2026	Yes	Systematic review (PRISMA, 40 papers) [22]	Architectural design education [22]	AI sparring as pedagogical framework [22]
Mr. Udit Gupta et al., 2025	Yes	Review / position paper [5]	Architecture [5]	AI ethics and collaboration in architecture [5]

Study	Full text retrieved?	Study type	Design domain	Primary focus
Stefanie Hutka, 2024	Abstract only	Framework development [23]	GenAI design (general) [23]	Problem-Solution Symbiosis framework [23]
Qian Huang & King Wang Poon, 2026	Abstract only	Qualitative thematic analysis (80 teams) [10]	Design education [10]	SuperSkillsStack framework for human-AI collaboration [10]
Gerry Derksen & Sejal Bansal, 2024	Yes	Course-based action research [12]	Visual communication / design education [12]	AI integration in design curriculum [12]
Wilmer Gonzalo Chaca Espinoza, 2025	Abstract only	Literature review (23 studies, 2016–2024) [7]	Industrial design [7]	Designer role transformation in Design Thinking ideation [7]
Thijs Willems et al., 2025	Yes	Pilot study (reflections from 500+ students) [11]	Engineering and architecture education [11]	Student agency and selective AI use [11]
Sonia Laugier, 2025	Abstract only	Theoretical essay [24]	Design (general) [24]	AI's effect on formal design languages [24]
Neil Leach, 2018	Abstract only	Essay / overview [25]	Design professions (general) [25]	AI's impact on design process and terminology [25]

The corpus is weighted toward reviews and conceptual papers, with empirical contributions coming from educational settings (course-based studies, student reflections, and design experiments). Full texts were available for 17 of the 25 sources; the remaining eight were analyzed from abstracts only. Design domains span architecture, graphic design, product and industrial design, textile design, interior design, interaction design, and cross-domain design thinking. Publication years cluster in 2024–2026, reflecting the rapid proliferation of generative AI scholarship, with two earlier essays (Wong, 2018; Leach, 2018) providing pre-generative-AI perspective.

Thematic Analysis

Defining Artificial Intelligence and Its Sub-fields

A foundational requirement for introducing AI to non-specialist readers is a clear definitional hierarchy. Across the corpus, AI is most frequently described as a family of computational techniques that approximate cognitive functions—learning, reasoning, and pattern recognition—through data-driven models [8]. It is also characterized as a domain within computer science that aims to understand and replicate human intelligence in computational systems [21]. One source offers a further categorical distinction: AI can be divided into machines that mimic human thought processes (cognitive modeling and the Turing test approach) and rational machines guided by laws of thought and a rational-agent approach [21].

The nested relationship among AI sub-fields is clarified most explicitly by Thoring et al., who define AI as an um-

brella term for machines that can perceive, synthesize, and infer information; machine learning as a sub-discipline that trains algorithms with examples to make predictions; deep learning as a subset of machine learning using artificial neural networks to mimic human brain learning; and generative AI as systems that use deep learning algorithms to generate creative outputs based on given inputs or prompts [1]. Bartlett and Camba add a complementary process-level explanation: image-generative AI models are trained on large datasets of images paired with textual descriptions and work through a process called diffusion—adding noise to data and then denoising it to generate images based on user prompts [2]. Xiong et al. offer a concise working definition of generative AI (GenAI) as a class of AI systems capable of autonomously generating new content such as images, texts, and design solutions based on learned data patterns [16].

Several sources position AI not merely as a technical capability but as a relational concept within design practice. Liu describes it as a collaborative intelligent agent that intervenes in spatial cognition and design judgment [3]. Wardi et al. frame AI through Reciprocal Human-Machine Learning (RHML), emphasizing dynamic two-way learning between human designers and AI systems [13]. Wong notes that AI is good at completing discrete tasks such as scheduling an appointment but incapable of more general intellectual tasks such as knowing when to reschedule [20], a distinction that remains relevant for calibrating non-specialist expectations.

Forms of AI Most Relevant to Design Practice

The sources collectively identify five principal categories of AI technology with direct bearing on design workflows: decision support, optimization, computer vision, natural language processing (NLP), and generative systems.

AI category	Design applications	Representative tools or examples	Key sources
Decision support	Data crunching, option generation, routine testing to support interpretation and strategic decisions [8]	AI-driven analysis of usability videos, interaction logs [8]; sentiment analysis for poster design [18]	Mirzaei et al. [8]; Hafez [18]; Wardi et al. [13]
Optimization	Virtual crash tests, stress analyses, multi-criteria trade-offs [8]; energy efficiency and sustainability analysis [5]; spatial layout generation [4]	Topology optimization [17]; Finch 3D for parametric layout [4]; Spacemaker AI [5]	Di Dio et al. [17]; Troshkin & Makukhin [4]; Gupta et al. [5]
Computer vision	Image segmentation, visual memory models, aesthetic analysis [18]; object recognition [20]	Adobe Sensei [18]; convolutional neural networks for image processing [18]	Hafez [18]; Wong [20]
Natural language processing	Synthesis of user narratives [8]; natural language CAD interaction [6]; prompt-based design interaction (“vibe-coding”) [4]	CADgpt (GPT-4 to Rhino3D) [6]; ChatGPT for architectural representation [3]	Mirzaei et al. [8]; Bourgault & J. E [6]; Troshkin & Makukhin [4]

AI category	Design applications	Representative tools or examples	Key sources
Generative systems (text, image, 3D)	Mood board creation, concept ideation, rapid prototyping visuals, rendered documentation [2, 9]; stylistic variation and layout generation [18]	DALL-E, Midjourney, Stable Diffusion, Adobe Firefly [2, 3, 22]; Autodesk Generative Design [5]; Zhitu AI [16]	Bartlett & Camba [2]; Liu [3]; Xiong et al. [16]; Gupta et al. [5]

Generative systems receive the most attention across the corpus, consistent with the post-2022 surge in large-model capabilities. However, the optimization and NLP categories are particularly salient for parametric customization: parametric systems like Finch 3D automate spatial layout generation and accelerate work with design variations [4], while NLP-based interfaces allow designers to interact through semantic descriptions rather than technical commands [4]. For assistive product development, decision-support systems that translate constraints for non-specialists are noted as making parametric customization more accessible [8], and tools like CADgpt lower language and technical barriers for inclusive education [6].

AI Integration Across Design Process Stages

The sources map AI contributions onto recognized design-process models, most frequently the double-diamond (discover–define–develop–deliver) framework. The pattern that emerges is one of stage-specific augmentation rather than wholesale automation.

Design stage	AI contributions	Evidence
Discovery / Empathy	Synthesizing large volumes of user narratives to surface latent needs [8]; generating diverse ideas and concepts [9]; brainstorming and information synthesis [10]	Mirzaei et al. [8]; Lai et al. [9]; Huang & Poon [10]
Define / Problem framing	Data analysis and user research providing insights, e.g., addressing “How might we” questions [9]; problem framing with human judgment interpreting contextual information [10]	Lai et al. [9]; Huang & Poon [10]

Design stage	AI contributions	Evidence
Develop / Ideation and prototyping	Rapid verification of ideas and exploration of multiple visual and compositional solutions [4]; reducing design fixation and expanding conceptual exploration [7]; generating variations through generative models [1]; rapid exploration and refinement of design solutions [9]; accelerated prototyping through sketch-to-mockup translation [11]	Troshkin & Makukhin [4]; Chaca Espinoza [7]; Thoring et al. [1]; Lai et al. [9]; Willems et al. [11]
Deliver / Evaluation and documentation	Generating concept documentation and rendered visuals [9]; AI-based building performance simulation [3]; virtual crash tests and stress analyses [8]; user testing and A/B feedback [12]	Lai et al. [9]; Liu [3]; Mirzaei et al. [8]; Derksen & Bansal [12]
Personalization / Customization	Data-driven personalization in industry [8]; AI-improved user experiences through personalization in service design [17]; participatory creation channels for users with varying expertise levels [16]	Mirzaei et al. [8]; Di Dio et al. [17]; Xiong et al. [16]

A consistent finding is that AI contributions are strongest in divergent phases—broadening the space of possibilities through rapid generation of alternatives—while convergent phases demand human judgment for selection, synthesis, and contextual interpretation [1, 8, 10]. Derksen and Bansal, working with design students, extended the double-diamond into a “triple diamond” model to accommodate iterative prompt refinement and user testing as distinct stages of divergence and convergence [12]. Laugier observes a more fundamental epistemological shift: generative AI moves the design process from an “écriture des formes” (writing of forms through informed drawing) to an “exercice d’un regard” (exercise of a gaze), reframing the designer’s cognitive engagement with form [24].

For parametric customization specifically, Finch 3D exemplifies how AI automates spatial layout generation, enabling students and practitioners to quickly generate and compare parametric variants [4]. Autodesk’s Generative Design and Spacemaker AI produce multiple design iterations based on given parameters, supporting optimization against structural, energy, and sustainability criteria [5]. The Zhitu AI platform demonstrates how GenAI-based tools can support rapid image generation and secondary editing tailored to enterprise-specific scenarios [16].

The Designer-AI Relationship: Assistive Tool, Not Autonomous Substitute

The most robust consensus across the entire corpus is that AI should function as an assistive tool rather than an autonomous replacement for human designers. Every source that addresses the designer-AI relationship explicitly or implicitly supports this position [1–3, 5, 7, 8, 11, 13–17, 21, 22]. Several distinct models of the human-AI relationship have been proposed.

The **hybrid intelligence** model positions the designer as a mediator between algorithmic precision and human needs, acting as what Di Dio et al. term a "sense maker" who infuses meaning and context into AI-generated outputs [17]. Mirzaei et al. recommend allocating divergent exploration primarily to AI while retaining human judgment for convergent selection and ethical decision-making [8]. Liu describes three evolving architectural roles: from form-makers to creative directors who curate prompts, to system integrators who orchestrate cross-platform workflows, to ethical supervisors responsible for contextual adequacy and social impact [3].

The **reciprocal learning** model, advanced by Wardi et al., frames the relationship through RHML in which dynamic, two-way learning between human designers and AI systems occurs, with the Polanyi Paradox (humans know more than they can tell) and Algorithm Aversion serving as key mediating constructs [13]. This model emphasizes that AI cannot access the tacit knowledge embedded in design expertise.

The **role-tagging** model from Willems et al. provides an educational taxonomy distinguishing AI as tool (instrumental assistant for efficiency), teammate (collaborative partner for complex problem-solving), or neither (deliberate non-use when human presence or empathy is required) [11]. Students in their 500-person course learned not only to harness AI for speed but also to reject its outputs and divert reclaimed time into deeper user research [11].

The **cognitive accelerator** framing from Huang and Poon is especially relevant to the assistive-tool thesis: generative AI functions primarily as a cognitive accelerator rather than a replacement for human creativity, with students consistently relying on human judgment to interpret contextual information and validate AI-generated outputs [10]. Domain knowledge derived from field observations enabled students to detect inaccuracies in AI suggestions [10].

Across these models, the designer competencies required for effective AI collaboration converge on several themes: prompt engineering and communication skills [1, 4, 16]; data literacy and bias auditing [8]; critical evaluation and interpretation of algorithmic outputs [7, 14]; ethical reasoning [3, 15]; and the human capacities of Agency, Domain Knowledge, Imagination, and Taste identified in the SuperSkillsStack framework [10]. Chaca Espinoza synthesizes this evolution by describing the industrial designer as no longer substituted by AI but redefined as an integrator of multiple intelligences, whose work requires new technical, ethical, and reflexive competencies [7].

Limitations, Risks, and the Need for Human Oversight

The literature identifies a consistent set of risks associated with AI in design, which can be organized into six categories.

Risk category	Manifestations in design	Sources identifying this risk
Bias	Training data can reinforce stereotypes, overrepresent certain demographics or styles, and produce non-inclusive solutions [2, 8]; algorithmic bias threatens fairness and inclusivity [5]; cultural biases in training models affect ideation [7]	Mirzaei et al. [8]; Bartlett & Camba [2]; Di Dio et al. [17]; Gupta et al. [5]; Chaca Espinoza [7]; Wong [20]

Risk category	Manifestations in design	Sources identifying this risk
Opacity / Explainability	Deep neural networks are opaque, making decisions difficult for humans to understand [1]; black-box recommendations face skepticism and compliance barriers [8]; lack of transparency hinders accountability [5]	Thoring et al. [1]; Mirzaei et al. [8]; Gupta et al. [5]; Liu [3]
Error and unreliability	ChatGPT can hallucinate facts and make reasoning errors [1]; moments of misfire where AI suggestions led to significant mistakes [11]; pain points in prompt formulation and asset generation [16]; domain knowledge helped detect inaccuracies in AI suggestions [10]	Thoring et al. [1]; Willems et al. [11]; Xiong et al. [16]; Huang & Poon [10]
Overreliance and skill atrophy	Overreliance can lead to creative atrophy and homogenization of output [8]; diminished professional skills and reduced creativity [18]; loss of human reasoning abilities [21]; stylistic homogenization and smoothing of individual traits [14]; feelings of "cheating" when using AI tools [12]	Mirzaei et al. [8]; Hafez [18]; Değirmenci [21]; Shabanova [14]; Derksen & Bansal [12]; Stanimirović et al. [22]
Authorship and originality	AI-generated designs may lack originality or uniqueness across users [2]; tensions related to creative appropriation and shared authorship [7]; loss of authorship clarity [5]; loss of uniqueness in design [9]	Bartlett & Camba [2]; Chaca Espinoza [7]; Gupta et al. [5]; Lai et al. [9]
Ethical and social concerns	Use of training data without artist consent [2]; privacy obligations in sensitive domains [8]; social manipulation and invasion of privacy [21]; environmental impact of high energy consumption [17]; standardized outputs that constrain innovation [13]	Bartlett & Camba [2]; Mirzaei et al. [8]; Değirmenci [21]; Di Dio et al. [17]; Wardi et al. [13]

Mitigation strategies recommended across the corpus include deploying AI primarily in divergent exploration phases while relying on human judgment in convergent selection [8]; maintaining veto power and transparent records of AI's role [8]; using diverse training data and ensuring algorithmic transparency [8]; requiring documentation of AI use in design projects [2]; developing AI literacy encompassing technical competence, ethical reasoning, critical eval-

uation, and adaptive learning [15]; and structured role-reflection protocols that reward selective non-use alongside productive use [11]. The concept of "hallucination fire-drills"—routines for recognizing and cross-checking unreliable AI outputs with expert advice—emerged organically among engineering and architecture students [11].

Educational Approaches to AI Literacy in Design

Several sources offer pedagogical models that are directly relevant to writing an accessible introductory section for non-specialist readers. Teo and Chew propose a layered framework for AI literacy in design intelligence comprising five strata: foundational knowledge (data, algorithms, models, computation), AI literacy competencies (technical, critical, ethical, adaptive), design thinking methods, human-AI collaboration practices, and outcome application domains [15]. The SuperSkillsStack framework identifies four key human competencies—Agency, Domain Knowledge, Imagination, and Taste—that must be cultivated alongside AI tools for effective collaboration [10]. Liu recommends a triadic pedagogical system of "algorithmic understanding–spatial control–ethical governance" for integrating AI into architectural education [3].

In practice-based education, Willems et al. introduced a three-way taxonomy (tool, teammate, neither) that enabled over 500 first-year students to develop metacognitive awareness about when AI should and should not be employed [11]. Troshkin and Makukhin demonstrated that comparing students' own spatial solutions with algorithmically generated alternatives enhanced critical thinking and improved design decision quality [4]. Stanimirović et al. propose "AI sparring" as a pedagogical framework in which AI serves not as a helper but as a provocateur, pushing students toward deeper critical reflection and stronger concept development [22]. Bartlett and Camba recommend that educators introduce AI as one tool among many, require documentation of AI use, ensure students can complete tasks without AI, and engage students in discussions of ethical issues including bias, artist theft, and copyright [2].

These educational frameworks collectively reinforce the view that accessible AI introductions should not merely explain what AI does technically but should simultaneously equip readers to think critically about when, why, and how AI is appropriately deployed in design—and when it should be deliberately set aside.

Synthesis

The 25 sources converge strongly on the fundamental characterization of AI as a family of computational techniques for pattern recognition, learning, and generation that augments rather than replaces human design intelligence. The definitional hierarchy—AI as umbrella term, machine learning as the training paradigm, deep learning as the neural-network implementation, and generative AI as the creative-output application layer [1]—is consistent across the sources that address it, providing a reliable scaffold for non-specialist explanation.

Where apparent divergence exists, it is attributable to differences in design domain and stage emphasis rather than genuine disagreement. Sources focused on architecture and spatial design emphasize optimization and parametric systems [3–5], while those focused on graphic and visual communication foreground image-generative tools [2, 12, 18]. Sources addressing early ideation highlight AI's capacity to widen creative search spaces and reduce design fixation [7, 8], while those addressing later stages emphasize simulation, evaluation, and documentation [3, 9]. These are complementary perspectives on the same underlying principle: AI's utility is stage-specific, and its value depends on the match between computational capability and the cognitive demands of a particular design task.

The tension between productivity gains and reflective depth—identified most explicitly by Stanimirović et al. [22] and Willems et al. [11]—offers a mechanistic explanation for why studies sometimes report mixed educational outcomes. When AI is deployed instrumentally as a production accelerator, students generate more outputs but may engage in shallower conceptual reasoning [22]. When AI is repositioned as a sparring partner or deliberately refused at certain stages, students develop stronger critical faculties and deeper design understanding [11, 22]. This pattern suggests

that the mode of AI integration—not merely its presence or absence—determines its impact on design quality and learning.

For parametric customization and assistive product development specifically, the literature supports a layered integration strategy. Optimization and parametric systems (e.g., Finch 3D, Autodesk Generative Design) handle constraint-based exploration and variant generation [4, 5]. NLP interfaces lower barriers for non-specialist users to interact with complex parametric tools through semantic rather than technical commands [4, 6]. Generative systems accelerate visual prototyping and concept communication [16]. Decision-support analytics inform trade-off analysis across competing design criteria [8]. In each case, the designer retains the irreducible responsibilities of problem definition, contextual interpretation, cultural sensitivity, and ethical judgment [3, 5, 7] —capacities that AI systems, as the corpus consistently demonstrates, do not possess.

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